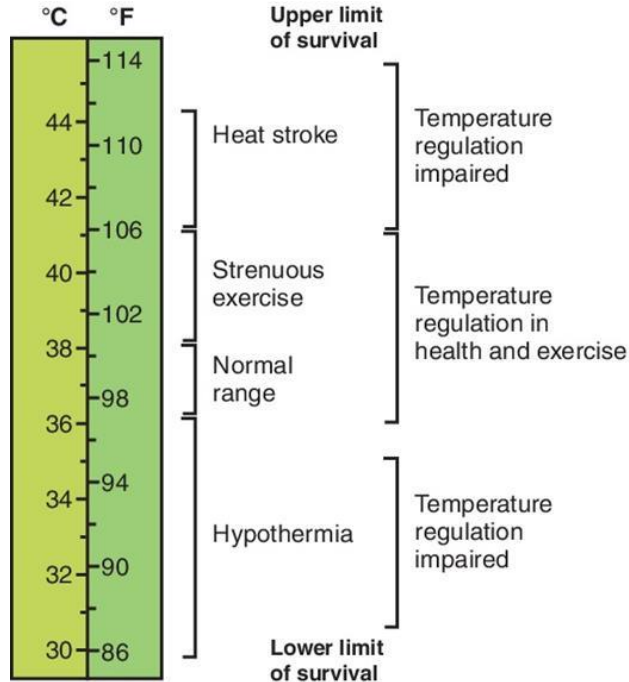


Session Objectives

At the conclusion of the session, the student should be able to:

1. Compare and contrast core body temperature and skin temperature under different conditions
2. Describe the homeostatic mechanisms of maintaining stable core body temperature, including heat production and dissipation
3. Understand how the central nervous system regulates body temperature
4. Describe the mechanism of fever

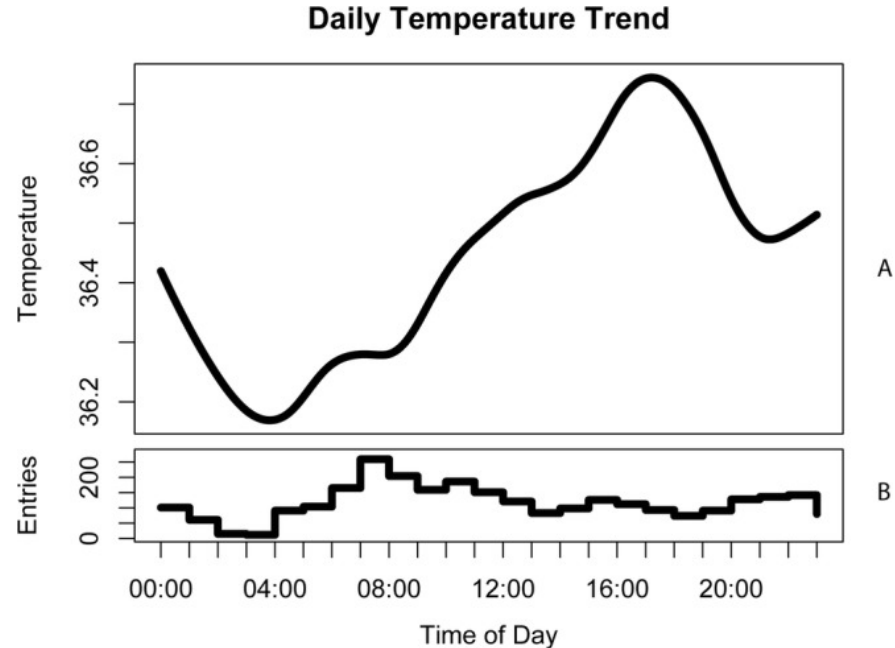
Survival ranges of body temperatures



- Internal body temperatures are regulated within a narrow range, despite wide variations in environmental temperature
- Failure of thermoregulatory mechanisms leads hyper- or hypothermia
- Extreme levels of hyper and hypothermia limit survival

Body temperature varies during the day

- Body temperature is lowest in the early morning and increases by 0.9-1.8°F by late afternoon
 - Minimum temperature - between 3 am and 5 am
 - Maximum temperature - between 4 pm and 6 pm
- A **morning** oral temperature of greater than **98.9°F** could be considered a fever, whereas a fever in the afternoon (and overall) is defined as **99.9°F**



JAMA 1992;268(12):1578-80.

J Gen Intern Med. 2018 Dec;33(12):2046-2047.

Using Smartphone Crowdsourcing to Redefine Normal and Febrile Temperatures in Adults: Results from the Feverprints Study

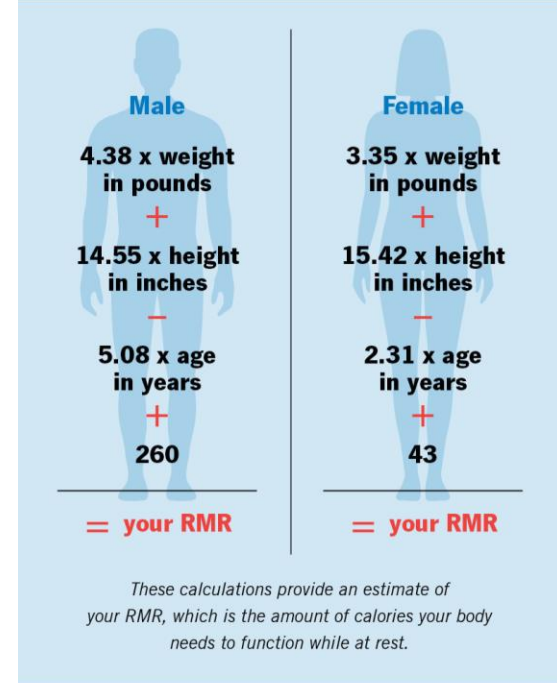
Basal/resting metabolic rate (BMR/RMR)

- Heat is produced as a byproduct of metabolism necessary to maintain chemical reaction in the brain and the internal organs
- BMR/RMR – metabolic rate under standard conditions at rest
 - 12 hour fast
 - Morning
 - 30 minutes rest
 - Comfortable temperature (~77F)
- BMR is measured by direct and indirect calorimetry
- BMR can be estimated based on the sex, weight, height, and age

Region	% of body mass	% of heat produced at rest at ambient temp.
Brain	2	16
Trunk viscera	34	56
Muscle and skin	56	18
Other	8	10

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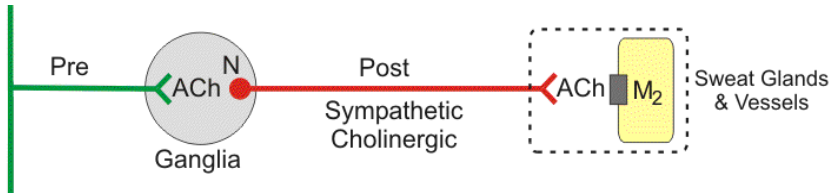
Heat production during external work/exercise

- Skeletal muscle becomes the main source of heat
- Metabolic rate can increase up to 10-fold
- Working muscles can warm circulating blood by 1-2°F

Region	% of body mass	% of heat produced at during exercise
Brain	2	1
Trunk viscera	34	8
Muscle and skin	56	90
Other	8	1

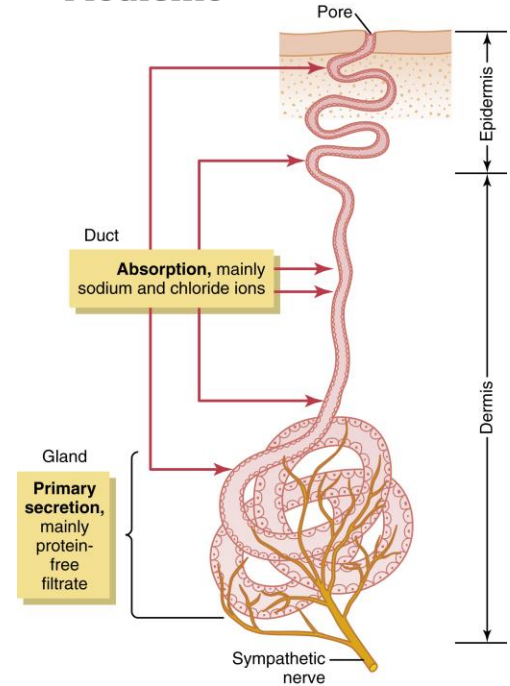
Mechanism of sweat production

- Eccrine glands of non-glabrous skin - face, trunk, and limbs
- Tubular structure consisting of two parts
 - deep subdermal coil
 - Produces primary secretion – composition of blood plasma minus the proteins
 - duct (in dermis and epidermis)
 - Reabsorbs sodium and chloride ions of the primary secretion
- The final composition of sweat depends on the rate of sweat production and flow, which is in turn is determine by the level of (cholinergic) sympathetic activation
- NOTE: Sweat glands are innervated by the post-ganglionic **cholinergic sympathetic** neurons



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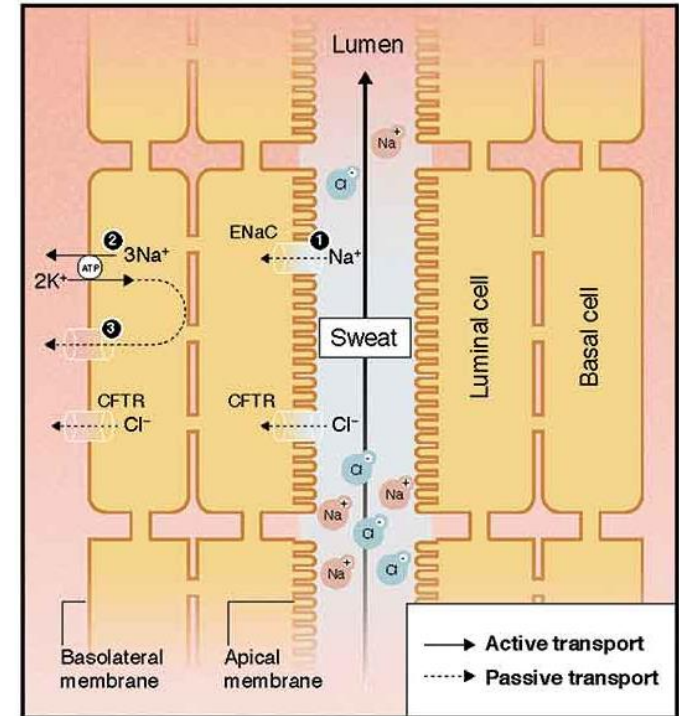
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Acclimation to heat—the role of aldosterone

- Excessive sweating leads to the reduction of plasma volume and activation of **aldosterone** production by the adrenal gland
- Aldosterone acts on the kidney
 - increases sodium and water reabsorption and increases plasma volume
- Aldosterone acts on the luminal cells of the sweat gland
 - upregulates epithelial sodium channel (ENaC), the first step in sodium reabsorption from the sweat

An unacclimated person can lose 15 to 30 grams of salt each day. After 4 to 6 weeks, the loss is 3 to 5 g/day



Effector Mechanisms

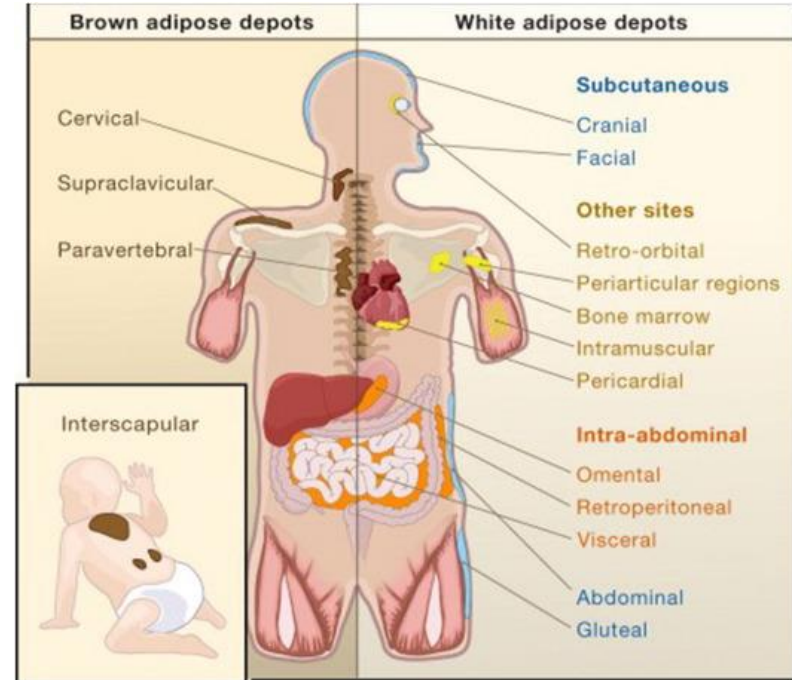
- Behavior
 - changing the environment
 - adding or removing clothing or other protective layering
- Temperature decreasing
 - Vasodilation of skin blood vessels
 - Sweating
- Temperature increasing
 - Vasoconstriction of skin blood vessels
 - Shivering
 - Chemical Thermogenesis

Chemical Thermogenesis

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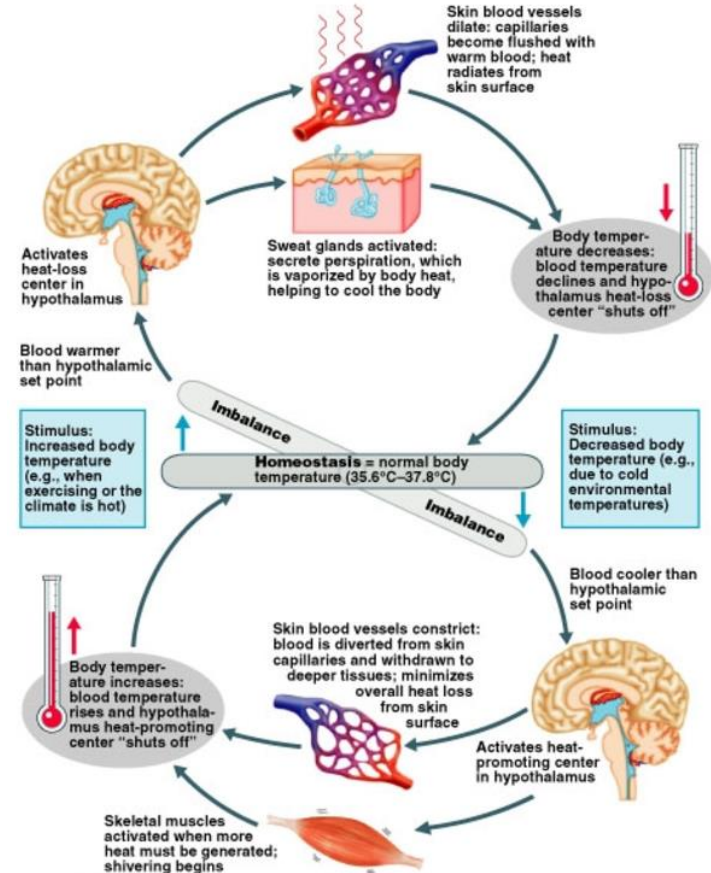
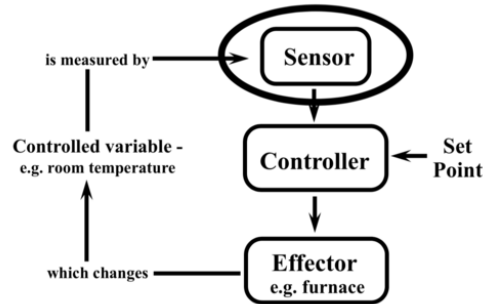
- Brown adipose tissue
- Abundant lipid droplets and mitochondria
- Sympathetic innervation
- Activates Lipolysis
- Mitochondria uncoupling protein 1 (UCP1)
- Uncoupling of electrochemical proton gradient and ATP synthesis
- Energy dissipates as heat



Temperature Set point

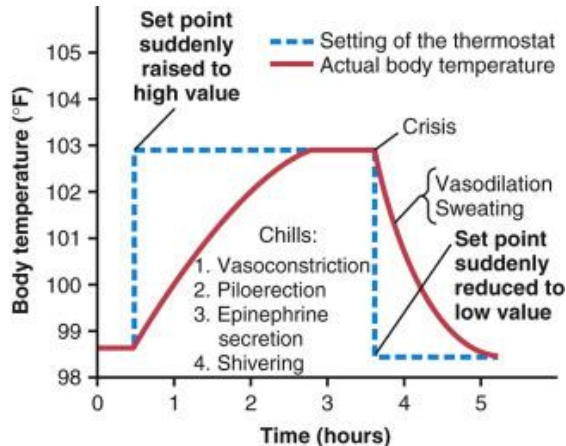
- Temperature set point is a homeostatic hypothalamic mechanism that activates heat production and dissipation to keep the core temperature constant
- Hypothalamic set point is often likened to a central room thermostat
- The set point can be affected by non-thermal signals/factors

- Exercise
- Ovulation
- Pyrogens

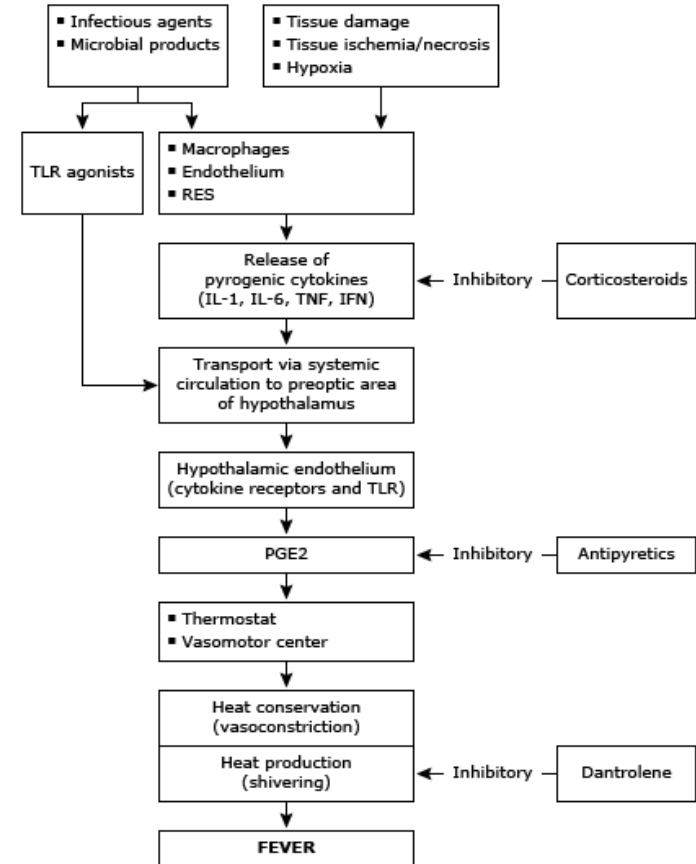


Mechanism of Fever

- Pyrogens – substances that increase the set point
- Symptomology
 - Chills** – onset of fever - the patient is responding to the elevated setpoint → cold skin (because of vasoconstriction) and shivering
 - Crisis** – resolution of fever - the factor that is causing the high set point is removed → set point drops to a normal level → cutaneous vasodilation and sweating



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Summary

- The core temperature remains stable
- Skin temperature fluctuates with the environment
- Heat dissipates by thermal radiation, evaporation of sweat, and conduction of heat to the ground and the air
- Cutaneous vasodilation increases heat transfer from the core to the body surface and helps heat dissipation
- The major sources of heat in the body are metabolism and normal physical activity
- Additional heat is produced from the food digestion and thermogenesis
- Thermoregulatory center in the hypothalamus responds to changes in skin and core temperature to activate or suppress neuronal and endocrine regulation of thermogenesis and heat dissipation
- Fever is defined as an increase in hypothalamic “set point” which results in the activation in heat production and reduction of heat dissipation